

Gaëlle Gilardot

Simulation Engineer | CFD · Multi-fidelity data · ML surrogates

+33 7 50 94 82 30 gaellegilardot@gmail.com Paris Portfolio Recommendation letter

SUMMARY CFD Engineer bridging physics-based simulation and machine learning, I have industry experience in liquid cooling for high-power chips, AI, automotive batteries and Formula 1. I turn simulations into decisions: I train ML surrogates (Python, OpenFOAM) and deliver measurable performance gains, and I am ready to bring that impact to the next challenge.

PROFESSIONAL EXPERIENCE

Associate Solutions Engineer - CDD | Corintis – Lausanne, Switzerland — May 2026 – Nov 2026

- Continuously built simulation and optimization tools to find the best-performing microfluidic cold plates for high-power chips.
- **Multi-fidelity benchmarking**: assessed 2.5D reduced-order solver accuracy against high-fidelity 3D CHT (OpenFOAM, Fidelity); bounded the error to define when ROM data can be trusted.
- Created a complete **AI-driven design optimization process** from an existing **CFD workflow** (Optuna-based): optimized Latin Hypercube seeding (**23× better design-space coverage**), **Gaussian-process Bayesian optimization** on parallel asynchronous workers, constraints and stopping criteria.
- Benchmarked Bayesian optimizers (GP), TPE & NSGA-II on Pareto hypervolume: **+10% hypervolume**, −1.5 °C on max temperature.
- Translated **customer specifications** into multi-objective constraints: **−1 to −2.9 °C** versus delivered customer designs.
- Developed **AI-assisted Python tools** for the engineering team:
  - **Adaptive power scaler** adjusting chip power maps to meet customer requirements with a new coolant.
  - **DoE pipeline** (meshing, parallel solves) grading manufacturing defects by thermal and pressure-drop impact.

Design & Development Engineer | Johnson Electric – Murten, Switzerland — Jan 2025 – Sep 2025

- Designed HVAC components in CAD and planned prototype test campaigns in line with regulations and test standards.
- Monitored experiments and built **Python scripts** to visualize and analyze large test datasets; detailed technical documentation.

Battery Calibration, CFD Simulation & Optimization Intern | AVL – Graz, Austria — Feb 2024 – Sep 2024

- Calibrated a Li-ion automotive battery model against test data by tuning its electrochemical equations during thermal runaway.
- Trained **RNN surrogate models** on simulation data and validated them (RMSE/NRMSE, overfitting control) to drive a **Python calibration tool** based on a 12-parameter DoE.
- Designed the CAD geometry and ran **CFD simulations** to feed the optimization loop; automated the full simulation workflow (setup, batch runs, post-processing) on a **Linux HPC cluster** and integrated the results into **data pipelines**.
- Wrote the internal technical documentation and co-authored a **scientific article** on the study publication.

Heat Exchanger Design & CFD Simulation Intern | AVL – Graz, Austria — Sep 2022 – Feb 2023

- Studied innovative **TPMS** (triply periodic minimal surface) heat exchangers for racing applications; led **CFD analyses** for a new Formula 1 engine cooling system, balancing thermal performance and aerodynamic constraints.

SKILLS

- **Programming & Tools:** Python (Optuna, scikit-learn, PyTorch, NumPy, pandas, SciPy) Git, MATLAB, Simulink, Java, Linux, HPC
- **Optimization & Machine Learning:** Gaussian-process surrogates, Bayesian optimization, RNN, TPE/NSGA-II, model validation (RMSE/NRMSE, overfitting), multi-objective & constrained optimization, Design of Experiments, LHS/Sobol sampling, AVL CAMEO
- **CFD & Thermal:** Fidelity, OpenFOAM, Ansys Fluent, Comsol, ParaView; conjugate heat transfer (CHT), k-ω SST turbulence
- **CAD & FEA:** ANSA, Siemens NX & Teamcenter, CATIA V5, SolidWorks

EDUCATION

MEng, Aeronautical & Mechanical Engineering | EPF Engineering School – Paris, France — 2019 – 2024

First Class Honours; ranked 1st of the 4th-year cohort.

High School Diploma, Engineering Sciences, with high honours, European section | Lycée Pierre Corneille — 2016 – 2019

ACADEMIC PROJECTS

- **Wing CFD validation against wind-tunnel data**: built a CFD model of an aircraft wing, prepared the wind-tunnel test model, and validated the CFD results against experimental data.
- **Satellite power modeling** (CNES, 4 months): MATLAB/Simulink model of battery charge and discharge with solar panels.

LANGUAGES & INTERESTS

Languages: French (native), English (fluent, C1, TOEIC 960), German (B2, working knowledge)

Interests: competitive badminton, swimming, classical dance

SOFT SKILLS

Learning agility, problem-solving and creativity, clear oral and written communication, adaptability and proactivity